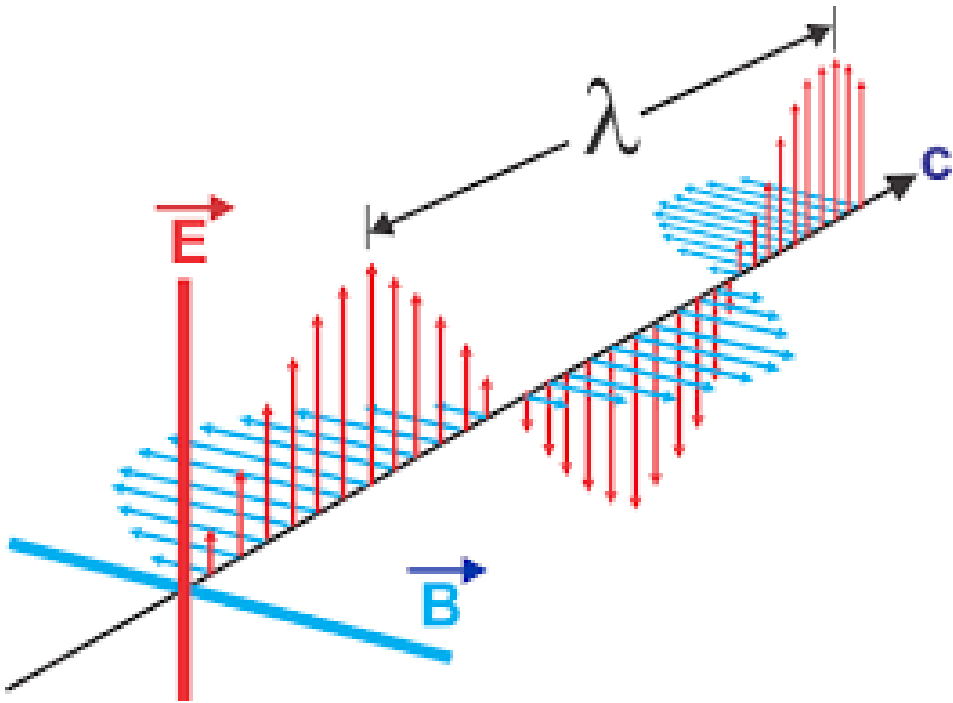


# EM Wave & Waveguides

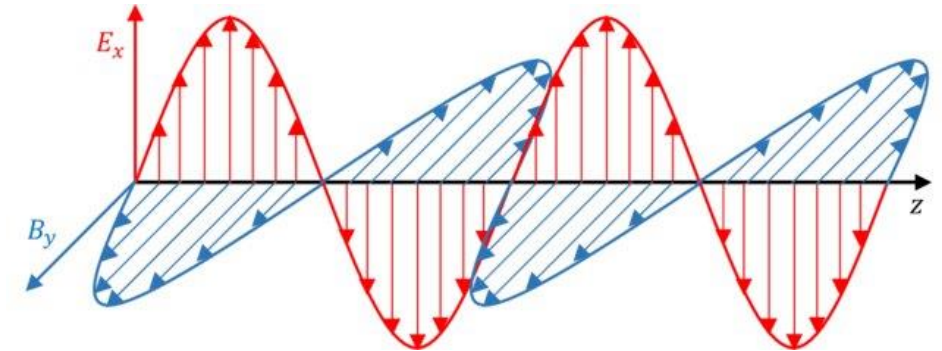
## Part-4



Ram Prakash  
Professor

Department of Physics  
Indian Institute of Technology (IIT), Jodhpur

Email: [ramprakash@iitj.ac.in](mailto:ramprakash@iitj.ac.in)



Background (Already discussed)

$$\left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left( \frac{\omega}{c} \right)^2 - k_z^2 \right] \widetilde{E}_{0z} = 0 \dots \dots \dots (24)$$

Similarly, we get

$$\left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left( \frac{\omega}{c} \right)^2 - k_z^2 \right] \widetilde{B}_{0z} = 0 \dots \dots \dots (25)$$

**Case-1** Longitudinal component of  $\vec{\widetilde{E}} = 0$

$\widetilde{E}_{0z} = 0$  but  $\widetilde{B}_{0z} \neq 0 \rightarrow$  transverse electric (TE) waves

**Case 2** Longitudinal component of  $\vec{\widetilde{B}} = 0$

$\widetilde{B}_{0z} = 0$  but  $\widetilde{E}_{0z} \neq 0 \rightarrow$  transverse magnetic (TM) waves

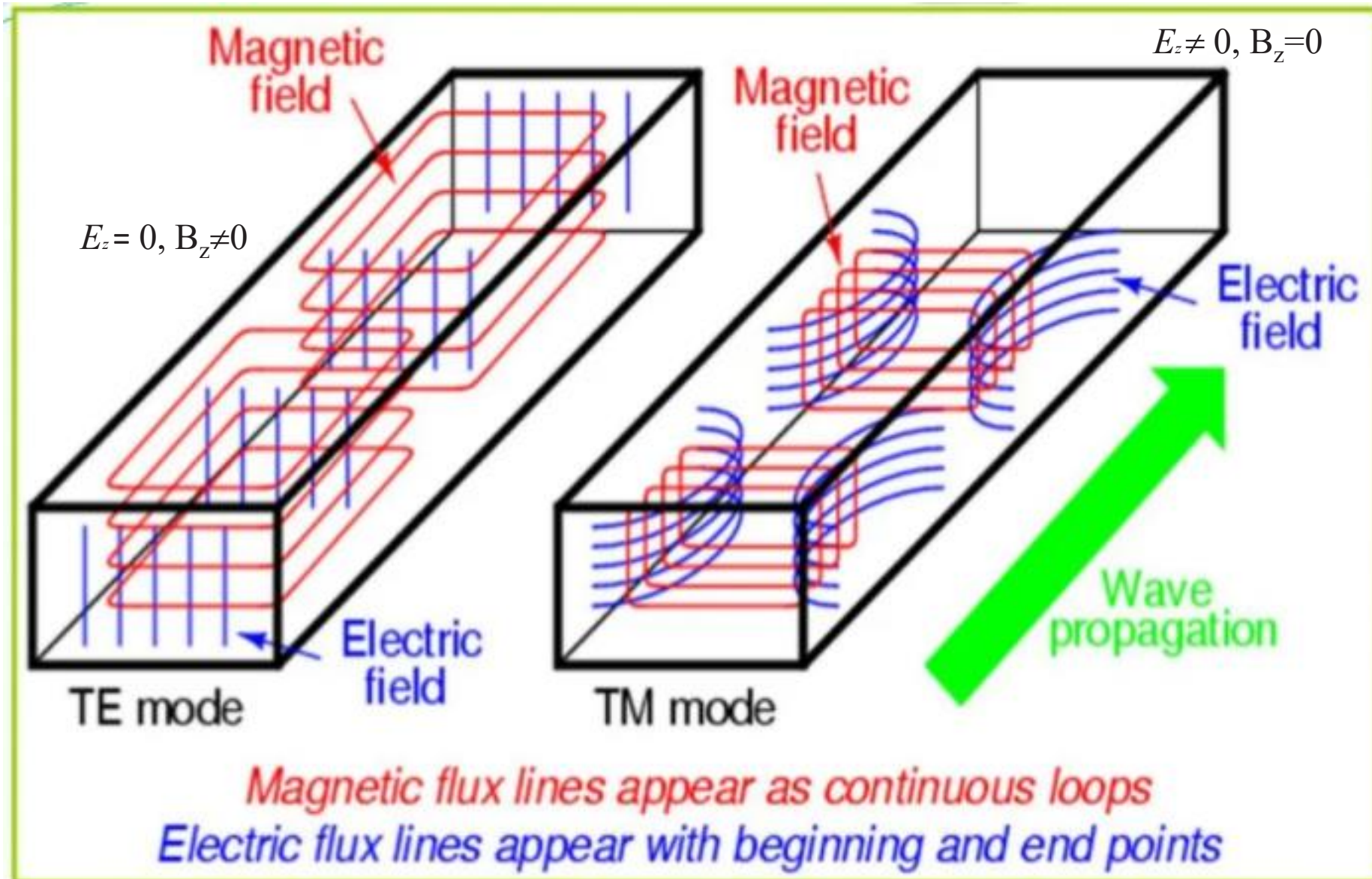
**Case 3** Transverse electric and magnetic waves

Both  $\widetilde{E}_{0z} = \widetilde{B}_{0z} = 0$

$\rightarrow$  no wave propagation, TEM Waves

Hence, transverse electric and magnetic waves cannot propagate in hollow waveguides.

# Basic features of waveguide



# TE Waves in a Rectangular Wave Guide

Consider a wave guide of rectangular shape, with height  $a$  and width  $b$ , and we are interested in the propagation of TE waves.

$E_z = 0$ , we call these **TE** (“transverse electric”)

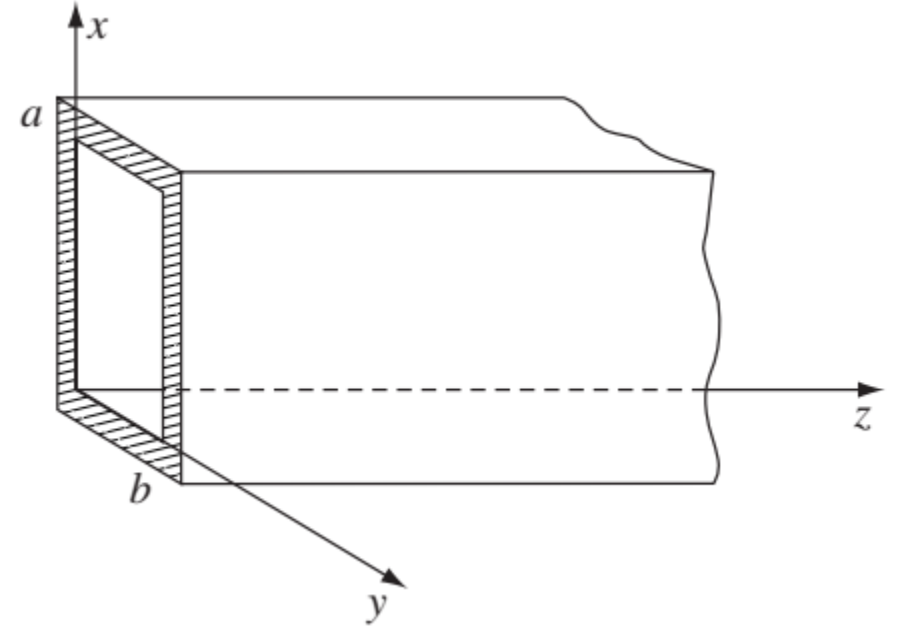
$$B_z(x, y) = X(x)Y(y), \quad (1)$$

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} + [(\omega/c)^2 - k^2] XY = 0.$$

Divide by  $XY$ , and note that the  $x$ - and  $y$ -dependent terms must be constants:

$$(i) \quad \frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2, \quad (ii) \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2,$$

$$\text{With} \quad -k_x^2 - k_y^2 + (\omega/c)^2 - k^2 = 0. \quad (2)$$



The general solution

$$X(x) = A \sin(k_x x) + B \cos(k_x x). \quad (3)$$

But the boundary conditions require that

for TE waves  $E_z = 0$  everywhere.

$dX/dx$ —vanishes at  $x = 0$  and  $x = a$ . So  $A = 0$ , and

Subjected to condition  $B^\perp = 0$  ( $B_x = B_y = 0$ )

This implies that

$$k_x = m\pi/a, \quad (m = 0, 1, 2, \dots). \quad (5)$$

$$\frac{\partial B_z}{\partial x} = \frac{\partial B_z}{\partial y} = 0$$

The same goes for  $Y$ , with

$$\text{Or} \quad \left. \frac{\partial B_z}{\partial n} \right|_s = 0$$

$$k_y = n\pi/b, \quad (n = 0, 1, 2, \dots), \quad (6)$$

where  $\mathbf{n}$  is the normal to the surface

and we conclude that

$$B_z = B_0 \cos(m\pi x/a) \cos(n\pi y/b).$$

This solution is called the  $\text{TE}_{mn}$  mode. (The first index is conventionally associated with the *larger* dimension, so we assume  $a \geq b$ . By the way, at least *one* of the indices must be nonzero—see Griffiths Book)

The wave number ( $k$ ) is obtained by putting Eqs. (5) and (6) into Eq. (2),

$$k = \sqrt{(\omega/c)^2 - \pi^2[(m/a)^2 + (n/b)^2]}.$$

If

$$\omega < c\pi\sqrt{(m/a)^2 + (n/b)^2} \equiv \omega_{mn},$$

the wave number is imaginary, and instead of a traveling wave we have exponentially attenuated fields. For this reason,  $\omega_{mn}$  is called the **cutoff frequency** for the mode in question. The *lowest* cutoff frequency for a given wave guide occurs for the mode  $TE_{10}$ :

$$\omega_{10} = c\pi/a. \quad \text{Or} \quad \omega_{1,0} = \frac{\pi}{\sqrt{\mu\epsilon} a}$$

Frequencies less than this will not propagate at all.

The wave number can be written more simply in terms of the cutoff frequency:

$$k = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2}.$$

The wave velocity (is greater than c)

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}},$$

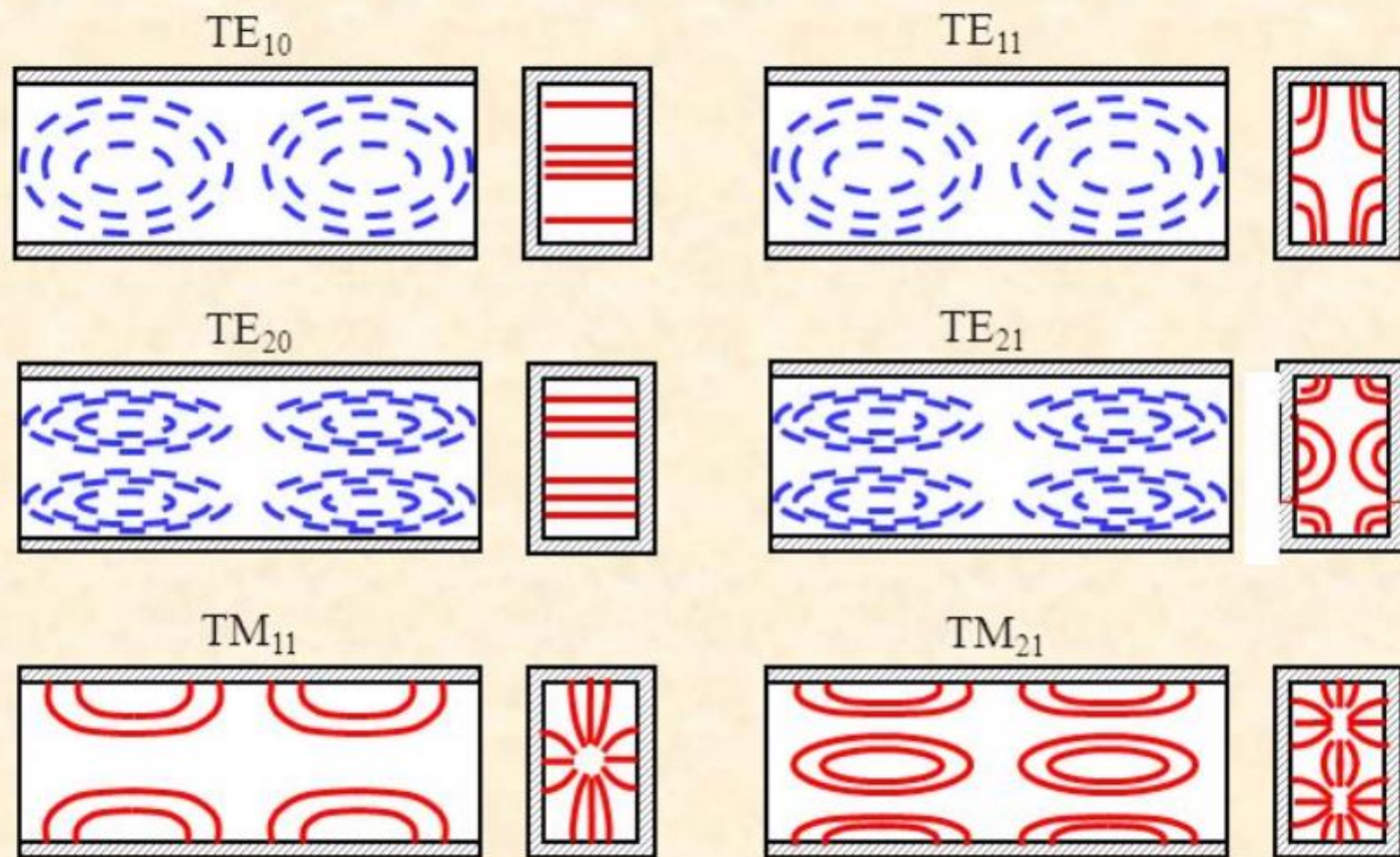
The energy carried by the wave travels at the *group* velocity

$$v_g = \frac{1}{dk/d\omega} = c\sqrt{1 - (\omega_{mn}/\omega)^2} < c.$$

The explicit fields for the mode denoted by  $\text{TE}_{1,0}$  are:

$$\begin{aligned} H_z &= H_0 \cos\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t} \\ H_x &= -\frac{ika}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t} \\ E_y &= i \frac{\omega a \mu}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t} \end{aligned}$$

## The modes with **higher** orders



— Electric field lines

- - - Magnetic field lines

# Key features of waveguide

- Waveguides are used to carry energy between pieces of equipment or over longer distances to carry transmitted power to an antenna or microwave signals from an antenna to receiver.
- Waveguides are made from copper, aluminum. The metals are extruded into long rectangular or circular pipes.
- An electromagnetic energy to be carried by a waveguide is injected in to one end of the waveguide.
- The electric and magnetic fields associated with the signal bounce off the inside walls back and forth as it progresses down the waveguide.

–



Rectangular waveguide



Waveguide to coax adapter



Waveguide bends



E-tee

# The Coaxial Transmission Line

A hollow wave guide cannot support TEM waves. But a coaxial transmission line, consisting of a long straight wire of radius  $a$ , surrounded by a cylindrical conducting sheath of radius  $b$ , *does* admit modes with  $E_z = 0$  and  $B_z = 0$ .



In this case Maxwell's equations yield

$$k = \omega/c$$

(so the waves travel at speed  $c$ , and are nondispersive),

$$cB_y = E_x \quad \text{and} \quad cB_x = -E_y$$

( $\mathbf{E}$  and  $\mathbf{B}$  are mutually perpendicular), and (together with  $\nabla \cdot \mathbf{E} = 0$ ,  $\nabla \cdot \mathbf{B} = 0$ )

$$\left. \begin{aligned} \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} &= 0, & \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= 0, \\ \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} &= 0, & \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= 0. \end{aligned} \right\} \quad (1)$$

These are the equations of *electrostatics* and *magnetostatics*, for empty space, in two dimensions.

Hence the solution with cylindrical symmetry can be borrowed directly from the case of an infinite line charge and an infinite straight current, respectively.

$$\mathbf{E}_0(s, \phi) = \frac{A}{s} \hat{\mathbf{s}}, \quad \mathbf{B}_0(s, \phi) = \frac{A}{cs} \hat{\boldsymbol{\phi}},$$

See Griffiths Book

for some constant  $A$ . Substituting these into

$$\left. \begin{aligned} \text{(i)} \quad \tilde{\mathbf{E}}(x, y, z, t) &= \tilde{\mathbf{E}}_0(x, y) e^{i(kz - \omega t)}, \\ \text{(ii)} \quad \tilde{\mathbf{B}}(x, y, z, t) &= \tilde{\mathbf{B}}_0(x, y) e^{i(kz - \omega t)}. \end{aligned} \right\}$$

and taking the real part:

$$\left. \begin{aligned} \mathbf{E}(s, \phi, z, t) &= \frac{A \cos(kz - \omega t)}{s} \hat{\mathbf{s}}, \\ \mathbf{B}(s, \phi, z, t) &= \frac{A \cos(kz - \omega t)}{cs} \hat{\boldsymbol{\phi}}. \end{aligned} \right\}$$

The coaxial cable is one of the common transmission example -cable TV. A typical experiment literally uses miles of transmission lines. The transmission line consists of an inner conductor – a wire – and an outer conductor – usually a metal braided jacket.



# The Coaxial Transmission Line:

## Geometrical Information

**a** = outside radius of inner conductor (inches)

**b** = inside radius of outer conductor (inches)

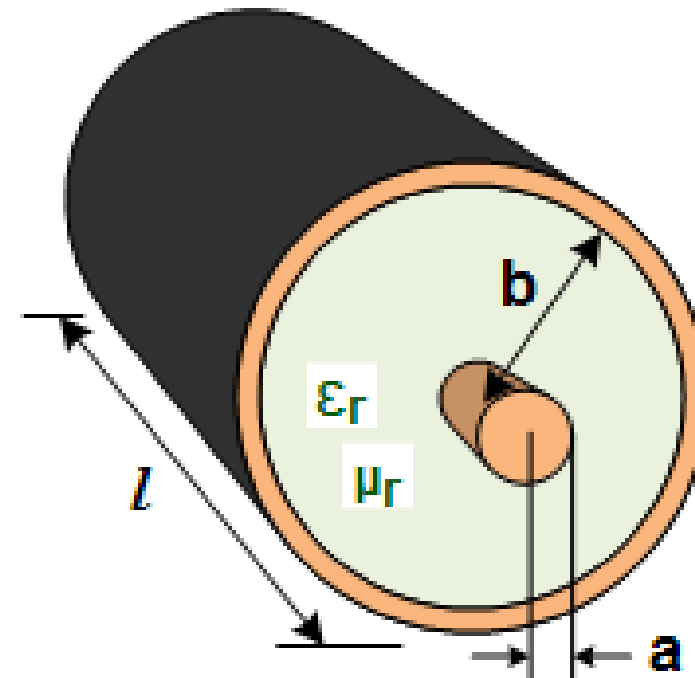
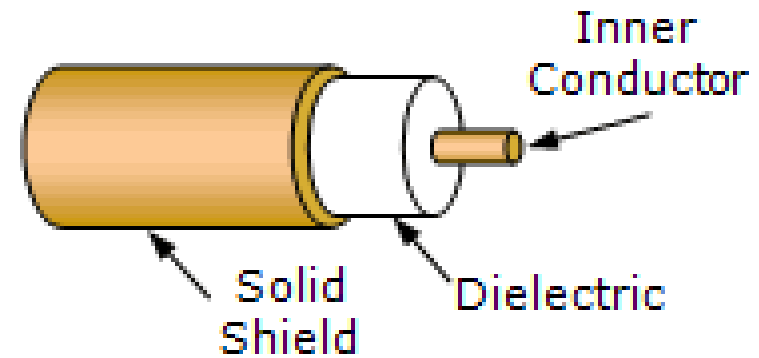
**c** = speed of light in a vacuum

**$\epsilon$**  = dielectric constant =  $\epsilon_0 * \epsilon_r$

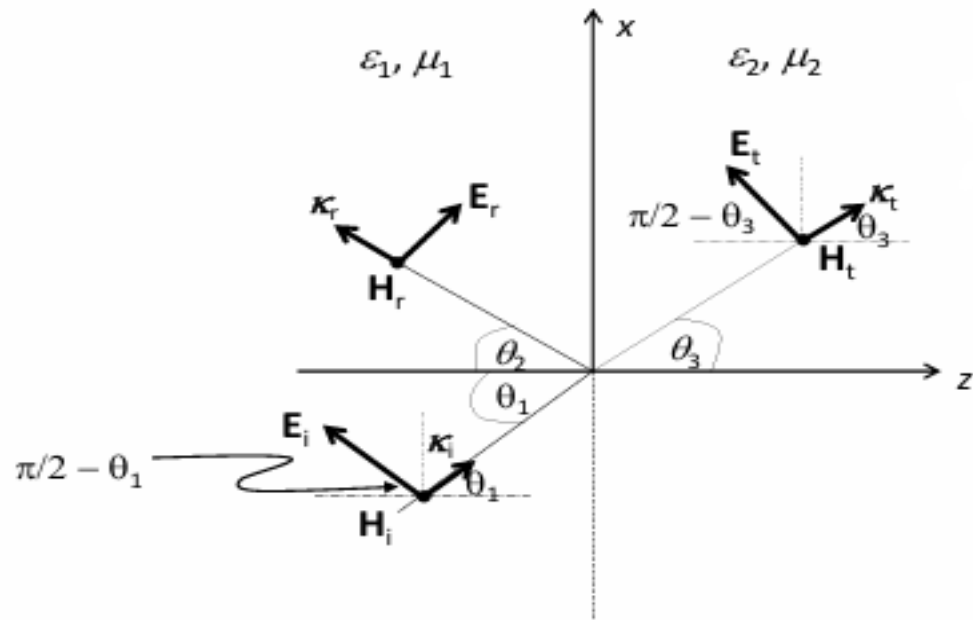
**$\epsilon_0$**  = permittivity of free space

**$\epsilon_r$**  = relative dielectric constant

**$\mu_r$**  = relative permeability



# Reflection and Transmission at the boundary between two dielectrics for oblique incidence



$$k_i = \omega \sqrt{\epsilon_1 \mu_1} (\hat{x} \sin \theta_1 + \hat{z} \cos \theta_1)$$

$$k_t = \omega \sqrt{\epsilon_2 \mu_2} (\hat{x} \sin \theta_3 + \hat{z} \cos \theta_3)$$

$$\vec{E}_i = \vec{E}_{1,0} e^{i(\omega t - \vec{k}_i \cdot \vec{r})}$$

$$\vec{E}_t = \vec{E}_{3,0} e^{i(\omega t - \vec{k}_t \cdot \vec{r})}$$

Simple case

Incident wave      Reflected wave      Transmitted wave

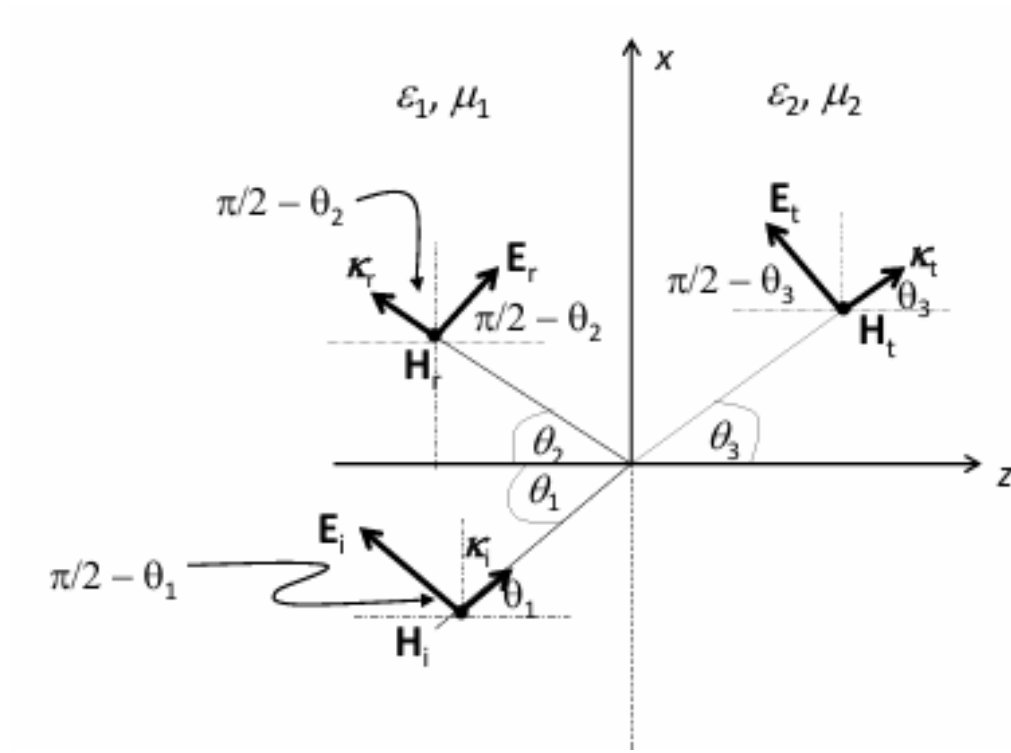
$$\begin{aligned} \vec{E}_i &= \hat{x} E_{10} e^{i(\omega t - k_1 z)} \\ \vec{E}_r &= \hat{x} E_{20} e^{i(\omega t + k_1 z)} \\ \vec{E}_t &= \hat{x} E_{30} e^{i(\omega t - k_2 z)} \end{aligned}$$

$$k_1 = \omega \sqrt{\mu_1 \epsilon_1} \quad \& \quad k_2 = \omega \sqrt{\mu_2 \epsilon_2}$$

where,

$$\vec{E}_{1,0} = E_{1,0} (\hat{x} \cos \theta_1 - \hat{z} \sin \theta_1)$$

$$\vec{E}_{3,0} = E_{3,0} (\hat{x} \cos \theta_3 - \hat{z} \sin \theta_3)$$



$$\vec{E}_r = \vec{E}_{2,0} e^{i(\omega t - \vec{k}_t \cdot \vec{r})}$$

where

$$\vec{E}_{2,0} = E_{2,0} ( \hat{x} \cos \theta_2 + \hat{z} \sin \theta_2 ) ;$$

$$k_r = \omega \sqrt{\epsilon_1 \mu_1} ( \hat{x} \sin \theta_2 - \hat{z} \cos \theta_2 )$$

From the boundary conditions :

Tangential component  $\Rightarrow$  x-component of  $\mathbf{E}$  at  $z = 0$  is continuous, it can be shown that

$$(\quad)e^{i(\omega t - \vec{k}_i \cdot \vec{r})} + (\quad)e^{i(\omega t - \vec{k}_r \cdot \vec{r})} = (\quad)e^{i(\omega t - \vec{k}_t \cdot \vec{r})}$$

$$k_i \sin \theta_1 = k_r \sin \theta_2 = k_t \sin \theta_3$$

$$\theta_1 = \theta_2 \quad : \text{Law of reflection}$$

Also,

$$n_1 k_0 \sin \theta_1 = n_2 k_0 \sin \theta_3 \quad k_0 = \omega/c$$

$$\Rightarrow n_1 \sin \theta_1 = n_2 \sin \theta_3 \quad : \text{Snell's Law}$$

From the boundary conditions for the tangential and normal components of E, H :

$$\begin{aligned} E_{1,0} \cos \theta_1 + E_{2,0} \cos \theta_1 &= E_{3,0} \cos \theta_3 \\ -\varepsilon_1 \sin \theta_1 E_{1,0} + \varepsilon_1 \sin \theta_1 E_{2,0} &= -\varepsilon_2 \sin \theta_3 E_{3,0} \end{aligned}$$

After some algebraic manipulations :

$$T = \frac{E_{3,0}}{E_{1,0}} = \frac{2 \varepsilon_1 \sin \theta_1 \cos \theta_1}{\varepsilon_2 \sin \theta_3 \cos \theta_1 + \varepsilon_2 \sin \theta_1 \cos \theta_3} : \text{transmitted field}$$

$$R = \frac{E_{2,0}}{E_{1,0}} = \frac{\varepsilon_1 \sin \theta_1 \cos \theta_3 - \sin \theta_3 \cos \theta_1}{\varepsilon_2 \sin \theta_3 \cos \theta_1 + \varepsilon_1 \sin \theta_1 \cos \theta_3} : \text{reflected field}$$

After further algebraic manipulations :

$$\frac{E_{2,0}}{E_{1,0}} = -\frac{\tan(\theta_1 - \theta_3)}{\tan(\theta_1 + \theta_3)}$$

$$\frac{E_{2,0}}{E_{1,0}} = -\frac{\tan(\theta_1 - \theta_3)}{\tan(\theta_1 + \theta_3)}$$

Observe that

1. For  $\theta_1 = \theta_3 \Rightarrow \tan(\theta_1 - \theta_3) = 0 \Rightarrow \text{No reflection} !$

Possible if  $n_1 = n_2 \Rightarrow$  two media are same i.e optically indistinguishable

2. For  $\theta_1 + \theta_3 = \pi/2 \Rightarrow \tan(\theta_1 - \theta_3) = \infty$   
 $\Rightarrow$  No reflection !

From Snell's Law we have

$$n_1 \sin \theta_1 = n_2 \sin \theta_3$$

$$\Rightarrow \text{For } \theta_3 = \frac{\pi}{2} - \theta_1$$

$$n_1 \sin \theta_{1p} = n_2 \cos \theta_{1p}$$

$$\Rightarrow \tan \theta_{1p} = \frac{n_2}{n_1} : \text{Brewster law}$$

↓

Brewster angle

Brewster's law states that when unpolarized light passes through a transparent substance at a specific angle, the light that is reflected is perfectly polarized

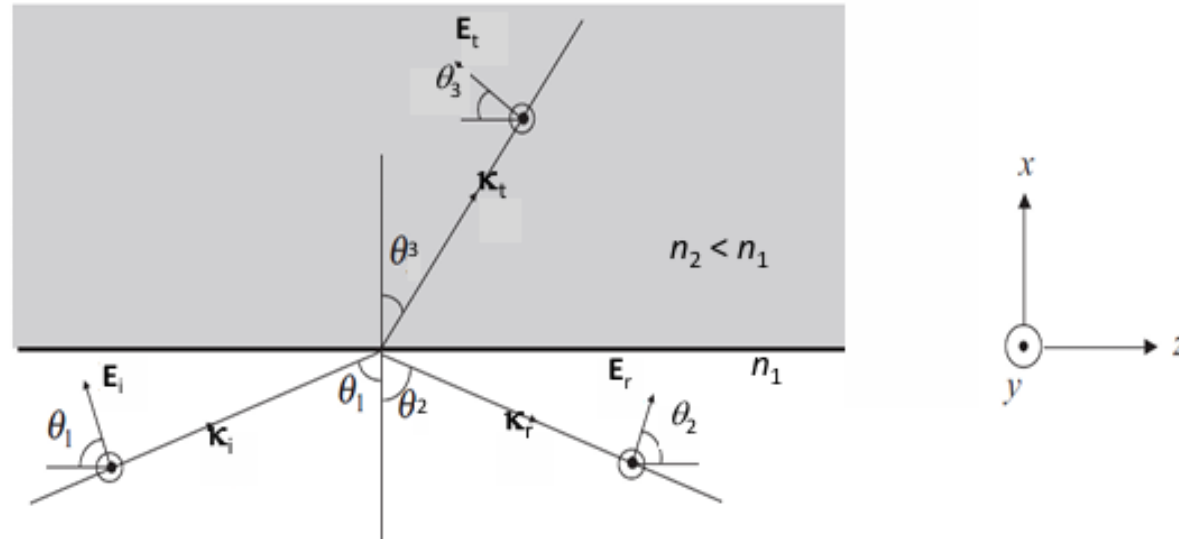
For glass and air :

$$n_2 = 1.5, n_1 = 1 \Rightarrow \theta_p = \tan^{-1}(1.5) = 56.3^\circ$$

Reflection and transmission of an em wave at an interface that separate two dielectrics  $\epsilon_{1,2}$

$$n_{1,2} = \sqrt{\epsilon_{1,2}/\epsilon_0}$$

Special Case :  $n_2 < n_1$



From Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_3 \Rightarrow \theta_3 > \theta_1$$

$$\sin \theta_3 = \frac{n_1}{n_2} \sin \theta_1 = \sqrt{(\epsilon_1/\epsilon_0)(\epsilon_2/\epsilon_0)} \sin \theta_1 \Rightarrow \sin \theta_3 = \sqrt{\epsilon_1/\epsilon_2} \sin \theta_1$$

Consider an angle  $\theta_1 = \theta_c = \textit{critical angle}$  for which  $\sin \theta_3 = 1$

$$\Rightarrow \theta_c = \sin^{-1}(n_2/n_1) = \sin^{-1} \sqrt{\epsilon_2/\epsilon_1}$$

$\Rightarrow$  For  $\theta_1 > \theta_c$ ,

$$\Rightarrow \sin \theta_3 > 1 !!$$

$\Rightarrow$  For  $\theta_1 > \theta_c$ ,

The incident wave returns to medium 1 of R.I.  $n_1$

$\Rightarrow$  Total internal reflection (TIR)

**Thanks**